

MA110: Lecture 18

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Real Quadratic Forms

Let $n \in \mathbb{N}$. A **real n -ary quadratic form** Q is a homogeneous polynomial of degree 2 in n variables with coefficients in $\mathbb{R}$. Thus

$$\begin{aligned} Q(x_1, \dots, x_n) &:= \sum_{j=1}^n \sum_{k=1}^n \alpha_{jk} x_j x_k \\ &= \sum_{j=1}^n \alpha_{jj} x_j^2 + \sum_{1 \leq j < k \leq n} (\alpha_{jk} + \alpha_{kj}) x_j x_k, \end{aligned}$$

where $\alpha_{jk} \in \mathbb{R}$ for $j, k = 1, \dots, n$.

Examples Let $a, b, c, a', b', c' \in \mathbb{R}$.

$n = 1 : Q(x) := ax^2$ (unary quadratic form)

$n = 2 : Q(x, y) := ax^2 + by^2 + a'xy$ (binary quadratic form)

$n = 3 : Q(x, y, z) := ax^2 + by^2 + cz^2 + a'xy + b'yz + c'zx$
(ternary quadratic form)

For $n \in \mathbb{N}$, consider an $n \times n$ real matrix $\mathbf{A} := [a_{jk}]$.

Then for $\mathbf{x} := [x_1 \ \cdots \ x_n]^\top$,

$$\begin{aligned}\mathbf{x}^\top \mathbf{A} \mathbf{x} &= [x_1 \ \cdots \ x_n] \begin{bmatrix} \sum_{k=1}^n a_{1k} x_k \\ \vdots \\ \sum_{k=1}^n a_{nk} x_k \end{bmatrix} = \sum_{j=1}^n \left(\sum_{k=1}^n a_{jk} x_k \right) x_j \\ &= \sum_{j=1}^n a_{jj} x_j^2 + \sum_{1 \leq j < k \leq n} (a_{jk} + a_{kj}) x_j x_k,\end{aligned}$$

which is an n -ary quadratic form.

In fact, $Q(x_1, \dots, x_n) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ for all $\mathbf{x} := [x_1 \ \cdots \ x_n]$ in $\mathbb{R}^{n \times 1}$ if and only if

$$\alpha_{jk} + \alpha_{kj} = a_{jk} + a_{kj} \quad \text{for all } j, k = 1, \dots, n.$$

In general, many $n \times n$ matrices induce the same quadratic form. For example, the matrices $\begin{bmatrix} 3 & 4 \\ 6 & 2 \end{bmatrix}$, $\begin{bmatrix} 3 & 5 \\ 5 & 2 \end{bmatrix}$, $\begin{bmatrix} 3 & -1 \\ 11 & 2 \end{bmatrix}$ induce the same binary quadratic form.

But if we require the matrix $\mathbf{A} := [a_{jk}]$ inducing the quadratic form Q to be symmetric, that is, $a_{jk} = a_{kj}$ for all j, k , then

$$a_{jk} = \frac{1}{2}(\alpha_{jk} + \alpha_{kj}) \quad \text{for all } j, k = 1, \dots, n.$$

Thus given an n -ary quadratic form Q , there is a unique $n \times n$ real symmetric matrix $\mathbf{A}$ such that $Q(x_1, \dots, x_n) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ for all $\mathbf{x} = [x_1 \ \dots \ x_n] \in \mathbb{R}^{n \times 1}$; in fact

$$\mathbf{A} := [a_{jk}], \quad \text{where } a_{jk} := \frac{1}{2}(\alpha_{jk} + \alpha_{kj}), \quad j, k = 1, \dots, n.$$

This real **symmetric** matrix $\mathbf{A}$ is called the **matrix associated with** the quadratic form Q , and we write $Q = Q_{\mathbf{A}}$.

A real n -ary quadratic form Q is said to be a **diagonal quadratic form** if there are $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ such that

$$Q(x_1, \dots, x_n) = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2.$$

It is clear that a quadratic form Q is diagonal if and only if Q is associated with a diagonal matrix $\mathbf{D}$, that is, $Q = Q_{\mathbf{D}}$.

Using the spectral theorem for real symmetric matrices, we show that every quadratic form can be orthogonally transformed to a diagonal quadratic form.

Theorem (Principle Axis Theorem)

Let Q be a real quadratic form and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be the symmetric matrix associated with Q . If $\mathbf{C}$ is an orthogonal matrix such that the matrix $\mathbf{D} := \mathbf{C}^T \mathbf{A} \mathbf{C}$ is diagonal, then $Q(\mathbf{x}) = Q_{\mathbf{D}}(\mathbf{y})$, where $\mathbf{y} := \mathbf{C}^T \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^{n \times 1}$.

Proof. Let $\mathbf{x} \in \mathbb{R}^{n \times 1}$ and $\mathbf{y} := \mathbf{C}^T \mathbf{x} = \mathbf{C}^{-1} \mathbf{x}$. Then $\mathbf{x} = \mathbf{C} \mathbf{y}$ and $Q_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = (\mathbf{C} \mathbf{y})^T \mathbf{A} (\mathbf{C} \mathbf{y}) = \mathbf{y}^T (\mathbf{C}^T \mathbf{A} \mathbf{C}) \mathbf{y} = \mathbf{y}^T \mathbf{D} \mathbf{y} = Q_{\mathbf{D}}(\mathbf{y})$.

To diagonalise a real n -ary quadratic form Q , we first write down the (real symmetric) matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ associated with Q . We then find an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ consisting of eigenvectors of $\mathbf{A}$ corresponding to its eigenvalues $\lambda_1, \dots, \lambda_n$ counted according to their algebraic multiplicities. If we let

$$\mathbf{C} := [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \quad \text{and} \quad \mathbf{D} := \text{diag}(\lambda_1, \dots, \lambda_n),$$

$$\text{Then } Q(\mathbf{x}) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2, \text{ where } \mathbf{y} := \mathbf{C}^T \mathbf{x} = \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \mathbf{x}.$$

Example

Let us transform the quadratic form

$Q(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3 - 4x_3x_1$ to a diagonal

form. Here $\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ is the associated matrix.

We have seen before that

$\mathbf{u}_1 := [1/\sqrt{2} \ 1/\sqrt{2} \ 0]^T$, $\mathbf{u}_2 := [-1/\sqrt{6} \ 1/\sqrt{6} \ 2/\sqrt{6}]^T$
and $\mathbf{u}_3 := [1/\sqrt{3} \ -1/\sqrt{3} \ 1/\sqrt{3}]^T$ are eigenvectors of $\mathbf{A}$
corresponding to the eigenvalues 3, 3 and -3 respectively, and
they form an orthonormal basis for $\mathbb{R}^{3 \times 1}$. Hence let

$$\mathbf{C} := \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \text{diag}(3, 3, -3).$$

Then $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$, and so $Q(\mathbf{x}) = 3(y_1^2 + y_2^2 - y_3^2)$, where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} := \mathbf{C}^T \mathbf{x} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

that is, $y_1 = (x_1 + x_2)/\sqrt{2}$, $y_2 = (-x_1 + x_2 + 2x_3)/\sqrt{6}$ and
 $y_3 = (x_1 - x_2 + x_3)/\sqrt{3}$.

Conic Sections

A **conic section** is the locus in $\mathbb{R}^2$ of an equation

$$a x^2 + b y^2 + c xy + a' x + b' y + c' = 0,$$

where $a, b, c, a', b', c' \in \mathbb{R}$ and at least one among a, b, c is nonzero. We assume WLOG that not both a and b are negative. It can be proved that the conic is one of these:

- (i) the empty set
- (ii) a single point
- (iii) one or two straight lines
- (iv) an ellipse
- (v) a hyperbola
- (vi) a parabola.

Terms of the second degree on the LHS of the equation give

$$Q(x, y) := a x^2 + b y^2 + c xy.$$

It is a binary quadratic form. It determines the type of the conic.

The (real symmetric) matrix associated with Q is

$$\mathbf{A} := \begin{bmatrix} a & c/2 \\ c/2 & b \end{bmatrix}.$$

Hence the equation of the given conic section becomes

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & c/2 \\ c/2 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a' & b' \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + c' = 0,$$

that is,

$$\begin{bmatrix} x & y \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y \end{bmatrix}^T + \begin{bmatrix} a' & b' \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix}^T + c' = 0.$$

Let $\mathbf{C} := [\mathbf{u}_1, \mathbf{u}_2]$ be an orthogonal matrix whose columns $\mathbf{u}_1$ and $\mathbf{u}_2$ are eigenvectors of $\mathbf{A}$ with corresponding eigenvalues λ_1 and λ_2 , and let $\mathbf{D} := \text{diag}(\lambda_1, \lambda_2)$ so that $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$.

We use the change of variables $\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix}$ to transform the quadratic form $Q(x, y)$ to a diagonal form as follows.

$$\begin{aligned}
Q(x, y) &= \begin{bmatrix} x & y \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y \end{bmatrix}^T \\
&= \begin{bmatrix} u & v \end{bmatrix} \mathbf{C}^T \mathbf{A} \mathbf{C} \begin{bmatrix} u & v \end{bmatrix}^T \\
&= \begin{bmatrix} u & v \end{bmatrix} \mathbf{D} \begin{bmatrix} u & v \end{bmatrix}^T \\
&= \lambda_1 u^2 + \lambda_2 v^2 = Q_D(u, v).
\end{aligned}$$

The ordered orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2)$ determines a new set of coordinate axes, so that the locus of the original equation is given by

$$\begin{aligned}
&\begin{bmatrix} u & v \end{bmatrix} \text{diag}(\lambda_1, \lambda_2) \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} a' & b' \end{bmatrix} \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix} + c' \\
&= \lambda_1 u^2 + \lambda_2 v^2 + \begin{bmatrix} a' & b' \end{bmatrix} [\mathbf{u}_1 \quad \mathbf{u}_2] \begin{bmatrix} u \\ v \end{bmatrix} + c' = 0.
\end{aligned}$$

If the conic so determined is not degenerate, that is, if it does not reduce to an empty set, a point, or line(s), then the signs of λ_1 and λ_2 determine the type of the conic section as follows.

The equation represents

1. **ellipse** if $\lambda_1 \lambda_2 > 0$, that is, both λ_1 and λ_2 are positive,
2. **hyperbola** if $\lambda_1 \lambda_2 < 0$, that is, one of λ_1, λ_2 is positive and the other is negative,
3. **parabola** if $\lambda_1 \lambda_2 = 0$, that is, one of λ_1, λ_2 is zero.

Note: Since $Q(x, y) := ax^2 + by^2 + cxy = \lambda_1 u^2 + \lambda_2 v^2$, where not both a and b are negative, it follows that not both λ_1 and λ_2 can be negative, and since the conic is assumed to be nondegenerate, not both λ_1 and λ_2 can be equal to zero.

Examples

1. Consider the conic section given by $2x^2 + 4xy + 5y^2 + 4x + 13y - 1/4 = 0$, and the binary quadratic form $Q(x, y) := 2x^2 + 4xy + 5y^2$.

Then the associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$ has eigenvalues $\lambda_1 := 1$ and $\lambda_2 := 6$, and the corresponding eigenvectors $\mathbf{u}_1 := [2 \ -1]^T / \sqrt{5}$ and $\mathbf{u}_2 := [1 \ 2]^T / \sqrt{5}$ form an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Hence let

$$\mathbf{C} := \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \text{diag}(1, 6).$$

Then $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$. So $Q(x, y) = Q_{\mathbf{D}}(u, v) = u^2 + 6v^2$, where

$$\begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \text{ that is,}$$

$$u = (2x - y)/\sqrt{5} \text{ and } v = (x + 2y)/\sqrt{5}.$$

Since $\begin{bmatrix} x \\ y \end{bmatrix} := \mathbf{C} \begin{bmatrix} u \\ v \end{bmatrix}$, substituting $x = (2u + v)/\sqrt{5}$ and $y = (-u + 2v)/\sqrt{5}$ in the given equation of the conic section, we obtain

$$u^2 + 6v^2 - \sqrt{5}u + 6\sqrt{5}v - \frac{1}{4} = 0.$$

Completing the squares, we see that

$$\left(u - \frac{\sqrt{5}}{2}\right)^2 + 6\left(v + \frac{\sqrt{5}}{2}\right)^2 = 9.$$

This is an equation of an [ellipse](#) with its centre at $(\sqrt{5}/2, -\sqrt{5}/2)$ in the uv -plane, where the u -axis and the v -axis are determined by the eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

2. Consider the conic section given by $2x^2 - 4xy - y^2 - 4x + 10y - 13 = 0$, and the binary quadratic form $Q(x, y) := 2x^2 - 4xy - y^2$.

The associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}$ has eigenvalues $\lambda_1 = 3, \lambda_2 = -2$, and the corresponding eigenvectors $\mathbf{u}_1 := [2 \ -1]^T / \sqrt{5}$ and $\mathbf{u}_2 := [1 \ 2]^T / \sqrt{5}$ form an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Hence let

$$\mathbf{C} := \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{D} := \text{diag}(3, -2).$$

Then $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$. Hence $Q(x, y) = Q_{\mathbf{D}}(u, v) = 3u^2 - 2v^2$,

$$\text{where } \begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix},$$

that is, $u = (2x - y)/\sqrt{5}$ and $v = (x + 2y)/\sqrt{5}$.

Thus substituting $x = (2u + v)/\sqrt{5}$ and $y = (-u + 2v)/\sqrt{5}$ in the given equation of the conic section, we obtain

$$3u^2 - 2v^2 - \frac{4}{\sqrt{5}}(2u + v) + \frac{10}{\sqrt{5}}(-u + 2v) - 13 = 0,$$

that is,

$$3u^2 - 2v^2 - \frac{1}{\sqrt{5}}(18u - 16v) - 13 = 0.$$

Completing the squares, we see that

$$\frac{(u - 3/\sqrt{5})^2}{4} - \frac{(v - 4/\sqrt{5})^2}{6} = 1.$$

This is an equation of a [hyperbola](#) with its centre $(3/\sqrt{5}, 4/\sqrt{5})$ in the uv -plane, where the u -axis and the v -axis are determined by the eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

3. Consider the conic section given by $9x^2 + 24xy + 16y^2 - 20x + 15y = 0$, and the binary quadratic form $Q(x, y) := 9x^2 + 24xy + 16y^2$. Then the associated symmetric matrix $\mathbf{A} := \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$ has eigenvalues $\lambda_1 := 25$ and $\lambda_2 := 0$, and the corresponding eigenvectors $\mathbf{u}_1 := [3 \ 4]^T/5$ and $\mathbf{u}_2 := [-4 \ 3]^T/5$ form an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Hence let $\mathbf{C} := \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$ and $\mathbf{D} := \text{diag}(25, 0)$. Then $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$. Thus $Q(x, y) = Q_{\mathbf{D}}(u, v) = 25u^2$, where $\begin{bmatrix} u \\ v \end{bmatrix} := \mathbf{C}^T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$, that is, $u = (3x + 4y)/5$ and $v = (-4x + 3y)/5$. Substituting $x = (3u - 4v)/5$ and $y = (4u + 3v)/5$ in the equation of the conic, we obtain $25u^2 + 25v = 0$, i.e., $u^2 = -v$, which is an equation of a **parabola** with its vertex at $(0, 0)$ in the uv -plane.

Quadric Surfaces

A **quadric surface** is the locus in $\mathbb{R}^3$ of an equation

$$a x^2 + b y^2 + c z^2 + a' xy + b' yz + c' zx + a'' x + b'' y + c'' z + d = 0,$$

where $a, b, c, a', b', c', a'', b'', c'', d \in \mathbb{R}$. We assume WLOG that not all three a, b and c are negative.

Terms of the second degree on the LHS of the equation give

$$Q(x, y, z) := a x^2 + b y^2 + c z^2 + a' xy + b' yz + c' zx.$$

It is a ternary quadratic form. It determines the type of the quadric surface. The real symmetric matrix associated with the quadratic form Q is

$$\mathbf{A} := \begin{bmatrix} a & a'/2 & c'/2 \\ a'/2 & b & b'/2 \\ c'/2 & b'/2 & c \end{bmatrix}.$$

Hence the equation of the given quadric surface becomes

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & a'/2 & c'/2 \\ a'/2 & b & b'/2 \\ c'/2 & b'/2 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} a'' & b'' & c'' \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + d = 0,$$

that is,

$$\begin{bmatrix} x & y & z \end{bmatrix} \mathbf{A} \begin{bmatrix} x & y & z \end{bmatrix}^T + \begin{bmatrix} a'' & b'' & c'' \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}^T + d = 0.$$

Let $\mathbf{C} := [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$ be an orthogonal matrix whose columns $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are eigenvectors of $\mathbf{A}$ with corresponding eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and let $\mathbf{D} := \text{diag}(\lambda_1, \lambda_2, \lambda_3)$.

We use the change of variables $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{C} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$ to transform

the quadratic form $Q(x, y, z)$ to a diagonal form as follows.

$$\begin{aligned}
Q(x, y, z) &= [x \ y \ z] \mathbf{A} [x \ y \ z]^T \\
&= [u \ v \ w] \mathbf{C}^T \mathbf{A} \mathbf{C} [u \ v \ w]^T \\
&= [u \ v \ w] \mathbf{D} [u \ v \ w]^T \\
&= \lambda_1 u^2 + \lambda_2 v^2 + \lambda_3 w^2 = Q_D(u, v, w).
\end{aligned}$$

The ordered orthonormal basis ($\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$) determines a new set of coordinate axes, so that the locus of the original equation is given by

$$[u \ v \ w] \text{diag}(\lambda_1, \lambda_2, \lambda_3) \begin{bmatrix} u \\ v \\ w \end{bmatrix} + [a'' \ b'' \ c''] \mathbf{C} \begin{bmatrix} u \\ v \\ w \end{bmatrix} + d = 0.$$

Leaving aside the degenerate cases, the primary cases are:

Equation	Surface	Eigenvalues of A
$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	ellipsoid	all three positive
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z}{c} = 0$	elliptic paraboloid	two positive, one zero
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$	elliptic cone	two positive, one negative
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	1-sheeted hyperboloid	two positive, one negative
$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	2-sheeted hyperboloid	one positive, two negative
$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z}{c} = 0$	hyperbolic paraboloid	one positive, one negative, one zero.

Pictures of these surfaces can be found on the Internet by searching with their names.

Example Consider the quadric surface given by

$$x^2 + y^2 + z^2 + 4xy + 4yz - 4zx - 27 = 0,$$

and the associated ternary quadratic form

$$Q(x, y, z) := x^2 + y^2 + z^2 + 4xy + 4yz - 4zx.$$

We have already transformed the associated symmetric matrix

$$\mathbf{A} := \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix} \text{ to a diagonal form, and have obtained}$$

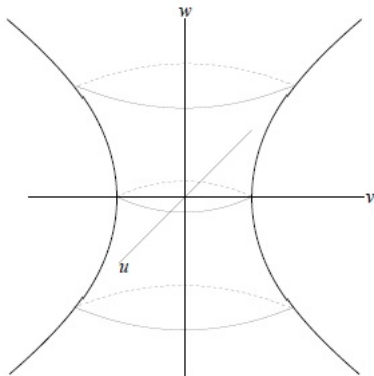
$$Q(x, y, z) = Q_{\mathbf{D}}(u, v, w) = 3(u^2 + v^2 - w^2) \text{ (with } x_1, x_2, x_3 \text{ and } y_1, y_2, y_3 \text{ in place of } x, y, z \text{ and } u, v, w),$$

where $\mathbf{D} := \text{diag}(3, 3, -3)$ and

$$u = (x + y)/\sqrt{2}, \quad v = (-x + y + 2z)/\sqrt{6}, \quad w = (x - y + z)/\sqrt{3}.$$

Under this change of coordinates, the quadric surface reduces to $u^2 + v^2 - w^2 = 9$.

This is an equation of a **one-sheeted hyperboloid** in the uvw -space, as shown in the following figure, where the u -axis, the v -axis and the w -axis are determined by the eigenvectors $\mathbf{u}_1 := [1/\sqrt{2} \ 1/\sqrt{2} \ 0]^T$, $\mathbf{u}_2 := [-1/\sqrt{6} \ 1/\sqrt{6} \ 2/\sqrt{6}]^T$ and $\mathbf{u}_3 := [1/\sqrt{3} \ -1/\sqrt{3} \ 1/\sqrt{3}]^T$. (See Lecture 16.)



Orthogonal Projection

Let $\mathbb{K}$ denote either $\mathbb{R}$ or $\mathbb{C}$. Recall that in Lecture 13, we have defined the (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n \times 1}$ in the direction of nonzero $\mathbf{y} \in \mathbb{K}^{n \times 1}$ as follows:

$$P_{\mathbf{y}}(\mathbf{x}) := \frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}.$$

In particular, if $\mathbf{y}$ is a unit vector, then $P_{\mathbf{y}}(\mathbf{x}) = \langle \mathbf{y}, \mathbf{x} \rangle \mathbf{y}$.

We noted that the vector $P_{\mathbf{y}}(\mathbf{x})$ is a scalar multiple of the vector $\mathbf{y}$, and proved the important relation

$$(\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})) \perp \mathbf{y}.$$

As a consequence,

$$\begin{aligned} \|\mathbf{x} - P_{\mathbf{y}}(\mathbf{x})\|^2 &= \langle \mathbf{x} - P_{\mathbf{y}}(\mathbf{x}), \mathbf{x} - P_{\mathbf{y}}(\mathbf{x}) \rangle \\ &= \langle \mathbf{x}, \mathbf{x} - P_{\mathbf{y}}(\mathbf{x}) \rangle \\ &= \|\mathbf{x}\|^2 - \langle \mathbf{x}, P_{\mathbf{y}}(\mathbf{x}) \rangle. \end{aligned}$$

More generally, let Y be a nonzero subspace of $\mathbb{K}^{n \times 1}$. We would like to find a (perpendicular) projection of $\mathbf{x} \in \mathbb{K}^{n \times 1}$ into Y , that is, we want to find $\mathbf{y} \in Y$ such that $(\mathbf{x} - \mathbf{y}) \in Y^\perp$. (This $\mathbf{y}$ is 'the foot of the perpendicular' from $\mathbf{x}$ into Y .)

If $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an orthonormal basis for the subspace Y , then a vector belongs to $Y^\perp$ if and only if it is orthogonal to each $\mathbf{u}_j$ for $j = 1, \dots, k$. As we saw while studying G-S OP, the vector

$$\tilde{\mathbf{y}} := \mathbf{x} - P_{\mathbf{u}_1}(\mathbf{x}) - \dots - P_{\mathbf{u}_k}(\mathbf{x}) = \mathbf{x} - \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 - \dots - \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k$$

is orthogonal to each $\mathbf{u}_j$ for $j = 1, \dots, k$, and so $\tilde{\mathbf{y}} \in Y^\perp$.

Since the vector $\mathbf{y} := \langle \mathbf{u}_1, \mathbf{x} \rangle \mathbf{u}_1 + \dots + \langle \mathbf{u}_k, \mathbf{x} \rangle \mathbf{u}_k$ belongs to Y , it is a (perpendicular) projection of $\mathbf{x}$ in Y .

The following result shows that this is the only vector in Y that works!